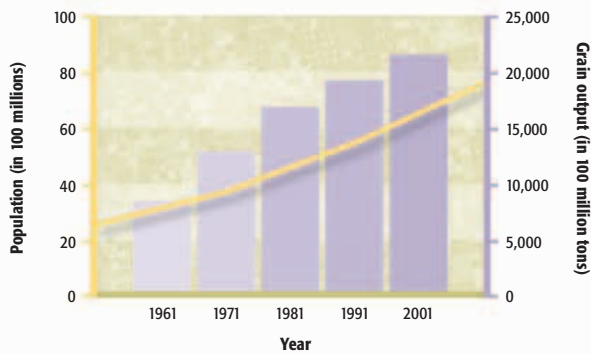




Rising demand

Despite fears that population growth would exceed our capacity to produce food, advances in farming methods have increased the productivity of the land. This graph charts world population against global grain output.

cruder forms, killed not only the targeted pests but also other insects and organisms that previously acted as natural inhibitors of the pest population. Pests develop resistance to chemical treatments and the same is true of plant diseases. It is not uncommon for new plant varieties to become obsolete in the space of three to four years, which is the time it takes for diseases and insects to adapt and destroy the crop. One serious effect of this problem is that the poorest farmers of the developing world are spending a

KEY

Population
Grain

large percentage of their income on new chemicals as they struggle to combat pest and disease resistance.

Food supply

The distribution of food aid to poorer countries came into practice to combat food shortages after World War II. The Indian famine of 1944 and China's great famine of 1959–61, where 30 million starved to death, triggered an international undertaking to establish secure and sustainable forms of food supply. Nations producing a food surplus pledged to maintain food supplies in the event of widespread crop failure. The US, the European Union (see pp.452–53), Canada, Japan, and Australia have

Delivering aid

Drought-stricken Niger, in northwest Africa, faced widespread famine in 2005. Thousands of children died before foreign aid was received.



been the biggest suppliers of food aid, but worldwide demand has often outstripped the ability to supply it.

Food production has matched the growth in worldwide population, but hunger and malnutrition remain a problem in some areas. This is especially the case in Africa, where many people still live in fear of hunger and starvation, and an estimated 200 million people suffer from malnutrition. The governments of many African

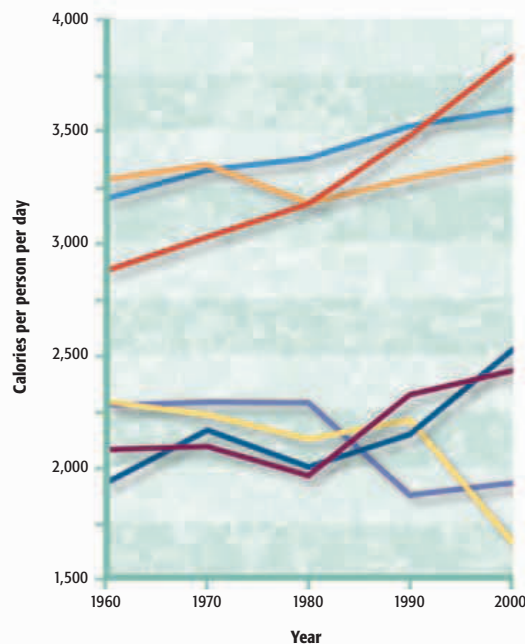


Rice research

Seedlings for "Golden Rice," a new rice variety modified to contain extra vitamin A, are grown in test tubes at an International Rice Research Institute laboratory.

countries are dependent on food aid. In 2003 climate conditions led to crop failure and famine in southern Africa and in 2006 droughts created crises in Ethiopia, Kenya, and Somalia—stark evidence that the inequalities that the Green Revolution sought to erase persist.

Rising oil prices and a switch to growing biofuels led to big rises in basic food prices from 2007, provoking the "tortilla riots" in Mexico in 2007 and violent protests in many African countries in 2008.



Food consumption

Although total food production has kept pace with the global rise in population (see above), the situation differs from country to country. Poorer nations have far less food available to their growing populations than more developed countries. This is reflected in the daily average caloric consumption (energy consumed from food) per person, as shown in this graph.

KEY

Central African Republic
Democratic Republic of Congo
France
India
United Kingdom
United States
Vietnam

AMERICAN SCIENTIST (1914–)

NORMAN BORLAUG

A key figure in the Green Revolution, Norman Borlaug was born into a farming family in Iowa. Borlaug studied forestry and microbiology before researching crop species and developing a new brand of high-yield wheat (see left). His goal was to feed the hungry people of the world. His groundbreaking work brought him many honors, including the Nobel Peace Prize in 1970.



While poorer regions are still dependent on one or two crops for survival, in the developed world an abundance and variety of food are available. Increased incomes have enabled an expanding middle class to adopt high-protein diets, and in the richest nations health risks associated with obesity are a growing problem. Many Western governments have felt compelled to introduce healthier eating initiatives to encourage the consumption of a balanced diet. Concerns have also been raised about the health risks created by chemical residues found in, or added to, produce. This has led to an increased demand for organic food, produced without the use of chemicals or artificial additives.

AFTER

Genetic modification is one of the newest methods of creating stronger crops.

GENETIC MODIFICATION

GMOs are organisms whose genetic material has been modified by adding extra material to generate new and useful **inherited traits**. Developed in the 1990s, the technique was applied to various crops, principally corn and soybeans, but also tomatoes, wheat, and many others, in the hope of **creating crops with better pest resistance**, disease resistance, shelf life, and taste. Concerns have arisen about the effects on **biodiversity** (the variety of species in a given ecosystem) that planting genetically-modified crops could have.



GM FRUIT